

ELENA CORTINA PINO '26 STATISTICS AND ECONOMICS ADVISOR: DR. IVAN RAMLER DEPARTMENT: STATISTICS



10th WORLD KUNG FU CHAMPIONSHIPS, 2025

- Kung fu is traditional wushu, a Chinese martial art rooted in the philosophical principles of **halting aggression**, and **moral and spiritual cultivation**.
- The World Kung Fu Championships (WKFC), hosted by the International Wushu Federation (IWUF), is an international level sporting event established in 2004 to propagate the development of wushu around the world.
- This competition welcomed more than 2100 athletes from over 50 nations competing in 73 kung fu events, across multiple age groups labelled A-F, scoring based on a 10-point judging system. This competition served as the basis for a pair of **Data Science educational modules** to be submitted to the **SCORE Network**.
- The variables chosen for this module are the **athletes' age groups**: A (11 years of age and below), B (12-14), C (15-17), D (18-39), E (40-59), F (60+ years old), and a selection of the **kung fu forms**.

SCORE NETWORK

- The SCORE Network is a NSF-funded collaborative initiative dedicated to creating and sharing Sports Content for Outreach, Research, and Education in statistics and data science.
- Following the required SCORE Network pedagogical and industry peer reviews, this educational module can become a published module in the SCORE Network Module Repository.

DATA AND NUANCES OF THE VARIABLES

- The data for this module comes from the Result Book of the 2025 10th World Kung Fu Championship, from The International Wushu Federation (IWUF).
- The Result Book is only available online as a 218-page pdf document containing the tables from each event, group and form separately. This pdf document was parsed into a csv document containing the same information.

CLUSTERING KUNG FU DATA FROM THE 10TH WORLD KUNG FU CHAMPIONSHIPS

CLEANUP PROCESS AND IMPUTATION

- Because of the competition's nature, not all of the kung fu forms had the exact same number of participants. This generated a significant amount of missing observations, especially for **groups A, B, and F**.
- Imputation** is a data replacement method where missing data is **substituted** with values to maintain dataset integrity, enabling complete analysis and reducing bias.
- The method selected for this case was using the **kNN** function from the **VIM** package in R.
- This process is the basis for a Data Cleaning module using the same principles used to process this data.

COMPLETE LINKAGE HIERARCHICAL CLUSTERING

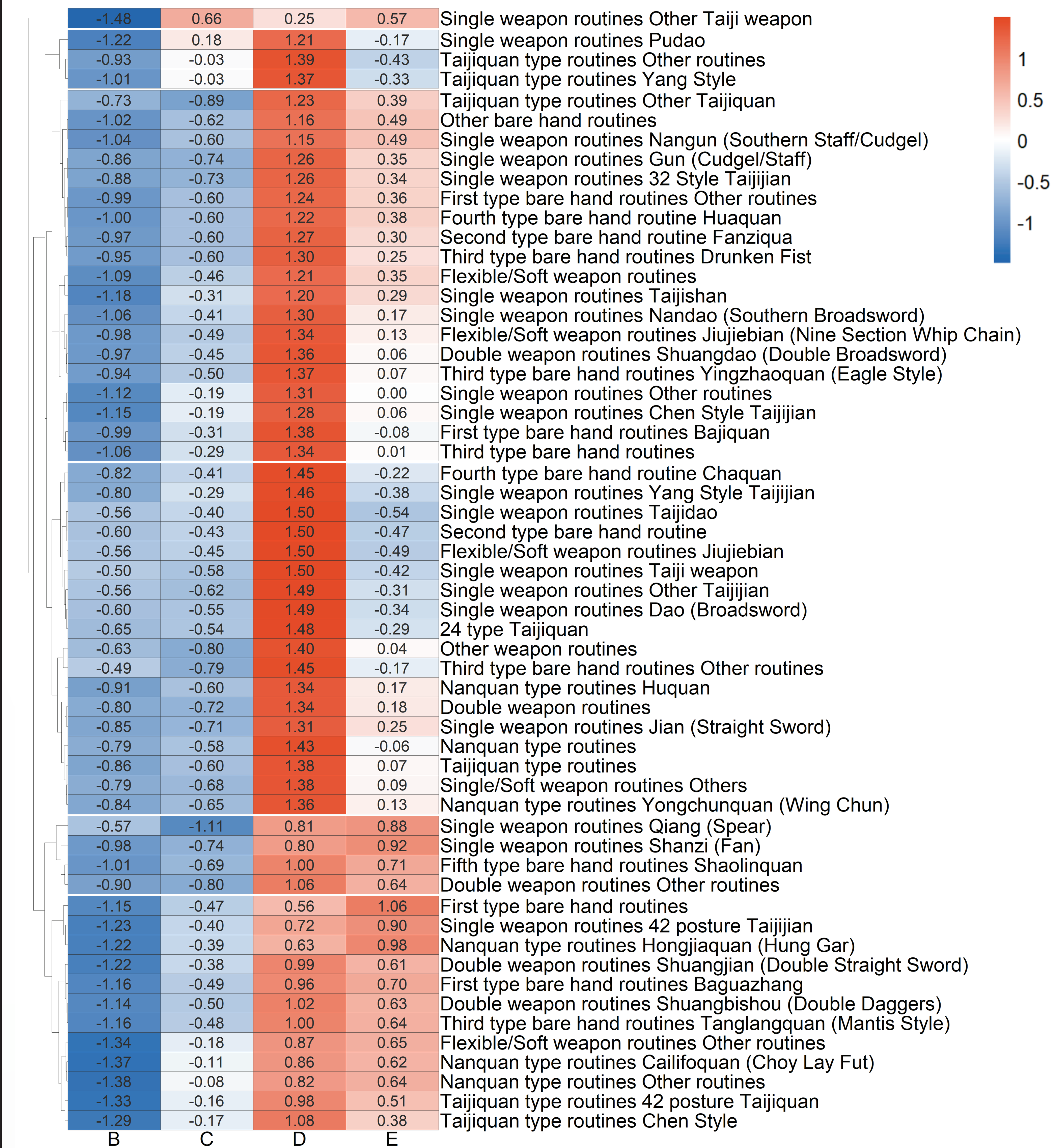
- Hierarchical clustering is a clustering method where the two **nearest** clusters are **repeatedly merged** until everything is merged into a **single cluster**.
- Complete (min-max) Linkage is a Machine Learning technique that groups clusters of data from the **furthestmost** point to the **point of reference**.
- The number of clusters was deliberately set to **6**. We could have opted for a different number of k-means clusters to be formed under the same mathematical principles. Using visual methods, such as looking at clear breaks in the **dendrograms**, is also a valid method to choose the number of clusters.
- Occasionally, one might have **relevant** subject-matter knowledge that gives a good guess of how many clusters we should have.

ROLE OF Z-SCORES FOR CLUSTERING

- The z-score tells how many **standard deviations** the value is **from the mean**, and is independent of the unit of measurement.
- We calculated z-scores for each **form** ensuring that each **row** of the data has **mean zero** and **standard deviation 1**. This allows a fair comparison across forms with varying difficulty by **isolating the age group effect**.
- Sample calculation for **Eagle Style** for **Group B's Z-score**
Mean of **Eagle Style Score Means** $\mu = 8.543$,
Group B Mean x = 8.398, **Standard deviation σ** = 0.154

$$z = \frac{8.398 - 8.543}{0.154} = -0.942$$

Complete Linkage Hierarchical Clustered Heatmap



RESULTS

- The bottom (6th) cluster is dominated by **non-weapon events**, abundantly **Taijiquan** events. These forms require **balance, technical ability and enhanced experience**. Power, explosivity nor athleticism warrant high scoring. **Groups E and F** have the **advantage**.
- The 5th cluster is dominated by 75% **weapon routines**. **Group B** scores **improve** and **C decreases** significantly from cluster 6.
- The 4th cluster's general patterns are **positive z-scores in E and sometimes F**, lower or moderate in group D, and negative in B. This cluster likely reflects **noise and stylistic diversity**. We are **not able to conclude that there's a clean technical grouping**. Abundance in "other"-forms creates **confounding groupings**.
- The forms in the 2nd and 3rd cluster require mainly **timing precision and expressive rhythm**. **Group B's** performance likely **lacks the refinement** to stand out.
- The 1st cluster reflects **noise** from by **Taiji weapon events**. The **technical clarity** and **precision** required to score high in **handling weapons** within Taiji makes **older athletes** excell.